A Quantum Network Distillation Protocol with the Ability of Link-parameter Reconstruction

***Abstract*—**
***Index Terms*—**

I. Introduction

Entanglement distillation and quantum network tomography(QNT) are two important components of quantum network. Entanglement distillation is important in quantum key distribution,

Quantum network tomography is important technique to estimate the each link-parameter of a quantum network. After determining the parameters of each link, we can estimate the enatanglement directly without some state tomography techniques, construct more reliabel network and select more effiencient entanglement distillation protocol just like the [1].

Entanglement distillation is an important technique

Entanglement distillation is essential for quantum key distribution , Unfortunately, repeated entanglement swapping operations needed for establishing long-distance communication exponentially degrade the end-to-end fidelity. Entanglement distillation protocols can be used to sup- press this fidelity loss by transforming a number of noisy entangled states into a smaller number of high- fidelity states using only local operations and classical communication

Existing QNT methods are mainly based on state tomography which requires local operations at monitors.

II. Preliminaries

A. Pauli–Liouville Representation

To facilitate the analysis of distillation, we present the Pauli–Liouville representation [2] in the two-qubit scenario below. Let $\mathbb{P}_2 \coloneqq \{I, X, Y, Z\}^{\otimes 2}$ denote the two-qubit Pauli operator basis, which is a complete orthogonal basis for the operator space $\mathcal{L}\big((\mathbb{C}^2)^{\otimes 2}\big)$ under the Hilbert–Schmidt inner product. Then, every two-qubit operator $\mathcal{G}$ can be uniquely expressed as a linear combination of the elements of $\mathbb{P}_2$. Defining the Pauli coefficient associated with $P \in \mathbb{P}_2$ as

$$s_P(\mathcal{G}) \coloneqq \mathrm{Tr}\big(P^\dagger \mathcal{G}\big)\,,$$

we can express $\mathcal{G}$ as follows,

$$\mathcal{G} = \frac{1}{4}\sum_{P\in\mathbb{P}_2} s_P(\mathcal{G}) P. \tag{1}$$

Following the Pauli-Liouville representation, we introduce the $|\cdot\rangle\!\rangle$ notation, and under it, the state can be expressed as a column vector as follows:

$$|\rho\rangle\!\rangle \coloneqq \big[s_P(\rho)\big]^T_{P\in\mathbb{P}_2}. \tag{2}$$

For a two-qubit linear map $\mathcal{E}$, its Pauli transfer matrix (PTM) $R_\mathcal{E}$ is a 16×16 matrix, whose enties are

$$(R_\mathcal{E})_{P,Q} \coloneqq \frac{1}{4}\,\mathrm{Tr}\big[P^\dagger \mathcal{E}(Q)\big]\,,\ \text{for any } P,Q\in\mathbb{P}_2. \tag{3}$$

Consequently, the outcome state of applying the operator $\mathcal{E}$ to an input state ρ can be expressed as follows,

$$|\mathcal{E}(\rho)\rangle\!\rangle = R_\mathcal{E}\,|\rho\rangle\!\rangle. \tag{4}$$

B. Pauli Channels in Bell-Diagonal Basis

Bell basis (composed by Bell states) is particularly convenient for describing two-qubit entanglement under Pauli noise because applying a single-qubit Pauli error to one half of a Bell pair maps it to another Bell state. The four Bell states are

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \tag{5}$$

Their density operators admit the Pauli decomposition

$$\begin{aligned}
|\Phi^+\rangle\langle\Phi^+| &= \frac{1}{4}\,(II + XX - YY + ZZ)\,,\\
|\Psi^+\rangle\langle\Psi^+| &= \frac{1}{4}\,(II + XX + YY - ZZ)\,,\\
|\Psi^-\rangle\langle\Psi^-| &= \frac{1}{4}\,(II - XX - YY - ZZ)\,,\\
|\Phi^-\rangle\langle\Phi^-| &= \frac{1}{4}\,(II - XX + YY + ZZ)\,.
\end{aligned} \tag{6}$$

These decompositions motivate the definition of the operator subspace

$$\mathcal{S}_\mathrm{B} \coloneqq \mathrm{span}\{II, XX, YY, ZZ\}. \tag{7}$$

An operator belongs to $\mathcal{S}_\mathrm{B}$ if it is a linear combination of these four Pauli operators.

A two-qubit density operator ρ is called Bell-diagonal if it is diagonal in the Bell basis. By Lemma 1, this is equivalent to $\rho \in \mathcal{S}_\mathrm{B}$.

For such a state, we define its restricted Pauli–Liouville vector as

$$|\rho\rangle\rangle_\mathrm{B} \coloneqq \begin{bmatrix} s_{II}(\rho) & s_{XX}(\rho) & s_{YY}(\rho) & s_{ZZ}(\rho)\end{bmatrix}^T, \tag{8}$$

where $s_P(\rho) \coloneqq \mathrm{Tr}(P^\dagger\rho)$.

The relation between the Bell basis and the Pauli basis is standard [3, 4]. For completeness, we state the corresponding change-of-basis relation below.

Lemma 1. *Consider the following ordered bases of $\mathcal{S}_B$:*

$$\begin{aligned}\mathcal{B}_{\text{Bell}} &\coloneqq \left(|\Phi^+\rangle\langle\Phi^+|, |\Psi^+\rangle\langle\Psi^+|, |\Psi^-\rangle\langle\Psi^-|, |\Phi^-\rangle\langle\Phi^-|\right), \\ \mathcal{B}_{\text{P}} &\coloneqq (II, XX, YY, ZZ).\end{aligned} \tag{9}$$

There exists a full-rank transformation matrix W which satisfies $4WW^T = I_4$ such that

$$\mathcal{B}_{\text{Bell}} = \mathcal{B}_{\text{P}} W, \qquad W = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix}. \tag{10}$$

For a normalized state ρ, its fidelity is defined as

$$F(\rho) = \frac{1}{4}\left[s_{II}(\rho) + s_{XX}(\rho) - s_{YY}(\rho) + s_{ZZ}(\rho)\right]. \tag{11}$$

Consider the single-qubit Pauli channel

$$\mathcal{P}(\sigma) = A\sigma + BX\sigma X + CY\sigma Y + DZ\sigma Z, \tag{12}$$

where $A, B, C, D \geq 0$ and $A + B + C + D = 1$. If the first qubit is unaffected and the second qubit undergoes $\mathcal{P}$, the corresponding two-qubit channel is $\mathcal{E} = \mathcal{I} \otimes \mathcal{P}$, where $\mathcal{I}$ denotes the identity channel. Applying this channel to the maximally entangled state $|\Phi^+\rangle$ produces the Bell-diagonal state

$$\begin{aligned}\rho &= \mathcal{E}\left(|\Phi^+\rangle\langle\Phi^+|\right) \\ &= A|\Phi^+\rangle\langle\Phi^+| + B|\Psi^+\rangle\langle\Psi^+| + C|\Psi^-\rangle\langle\Psi^-| + D|\Phi^-\rangle\langle\Phi^-|.\end{aligned} \tag{13}$$

As defined in (3), the full PTM $R_{\mathcal{E}}$ is a 16×16 matrix with respect to the complete two-qubit Pauli basis. However, $\mathcal{S}_{\text{B}}$ is invariant under $\mathcal{E}$, meaning that $\mathcal{E}(\mathcal{G}) \in \mathcal{S}_{\text{B}}$ for every $\mathcal{G} \in \mathcal{S}_{\text{B}}$. Therefore, the action of $\mathcal{E}$ on Bell-diagonal states is completely described by its restriction to $\mathcal{S}_B$. With respect to the ordered basis $\mathcal{B}_{\text{P}}$ in 9, the resulting 4×4 PTM is

$$\begin{aligned} R_{\mathcal{E}}|_{\mathcal{S}_B} &= \text{diag}(1, q_X, q_Y, q_Z), \\ q_X &= A + B - C - D = 2(A + B) - 1, \\ q_Y &= A - B + C - D = 2(A + C) - 1, \\ q_Z &= A - B - C + D = 2(A + D) - 1. \end{aligned} \tag{14}$$

Here, q_X, q_Y, q_Z are also the eigenvalues of the single-qubit Pauli channel on X, Y, Z, respectively. It is obvious that $q_X, q_Y, q_Z \in [-1, 1]$, which leads to the identification challenge in QNT, elaborated later in §IV-A.

Consequently, the output state in (13) has the Pauli decomposition

$$\rho = \frac{1}{4}\left(II + q_X XX - q_Y YY + q_Z ZZ\right), \tag{15}$$

and its restricted Pauli–Liouville vector is

$$|\rho\rangle\rangle_{\text{B}} = \begin{bmatrix} 1 & q_X & -q_Y & q_Z \end{bmatrix}^T. \tag{16}$$

By Lemma 1, the Pauli coefficients of state ρ uniquely determine the channel parameters A, B, C, D.

C. Double-Selection Protocol

Entanglement purification aims to transform multiple copies of a noisy entangled state into fewer pairs with higher fidelity to a target Bell state, which we take to be $|\Phi^+\rangle$. For example, the protocol considered here uses two noisy ancillary pairs to probabilistically increase the fidelity of a third pair. Recent work has shown that measurement statistics collected during entanglement distillation can also be used for state estimation [5].

We consider the double-selection protocol introduced by Fujii and Yamamoto [6]. Figure 1 illustrates its dual role: successful attempts yield purified entangled pairs, while measurement records from all attempts are used to estimate the input Bell-diagonal state under the conditions specified below.

Alice and Bob initially share three independent copies of the Bell-diagonal state [XW: (called *source state*)] in (15) through a noisy quantum channel. Denote the copies by ρ_1, ρ_2, ρ_3, with $\rho_1 = \rho_2 = \rho_3 = \rho$.

The protocol proceeds as follows.

1) Alice and Bob each locally apply a CNOT gate, using their qubit from ρ_1 as the control and their qubit from ρ_2 as the target.
2) They each locally apply a second local CNOT gate, using their qubit from ρ_3 as the control and their qubit from ρ_2 as the target.
3) They measure ρ_2 in the Z basis and ρ_3 in the X basis. Let z_A, z_B and x_A, x_B denote the local outcomes of Alice and Bob, respectively.
4) They retain ρ_1 if and only if $z_A = z_B$ and $x_A = x_B$; otherwise, they discard it.

We repeat the double-selection circuit and comparing Alice's and Bob's measurement outcomes yields the marginal and joint coincidence probabilities

$$\begin{aligned} p_Z &\coloneqq \Pr(z_A = z_B) = \frac{1 + q_Z^3}{2}, \\ p_X &\coloneqq \Pr(x_A = x_B) = \frac{1 + q_X^2}{2}, \\ p_{XZ} &\coloneqq \Pr(z_A = z_B,\ x_A = x_B) = \frac{1 + q_Z^3 + q_X^2 + q_Z q_Y^2}{4}. \end{aligned} \tag{17}$$

Here, p_{XZ} is also the acceptance probability of the protocol.

Casapao et al. [5] proved the following Lemma 2.

Lemma 2 (Reconstruction from double-selection statistics [5, [XW: Add where the results in the paper here]])**.** *Let ρ be the Bell-diagonal state in* (15)*. If $F(\rho) > 1/2$, then, under ideal local operations and measurements, the coincidence probabilities p_X, p_Z, p_{XZ} uniquely determine the nontrivial PTM eigenvalues q_X, q_Y, q_Z. Specially, from rearranging* (17)*, we have*

$$\begin{aligned} q_Z &= \sqrt[3]{2p_Z - 1}, \\ q_X &= \sqrt{2p_X - 1}, \\ q_Y &= \sqrt{\frac{4p_{XZ} - 2p_X - 2p_Z + 1}{\sqrt[3]{2p_Z - 1}}}. \end{aligned} \tag{18}$$

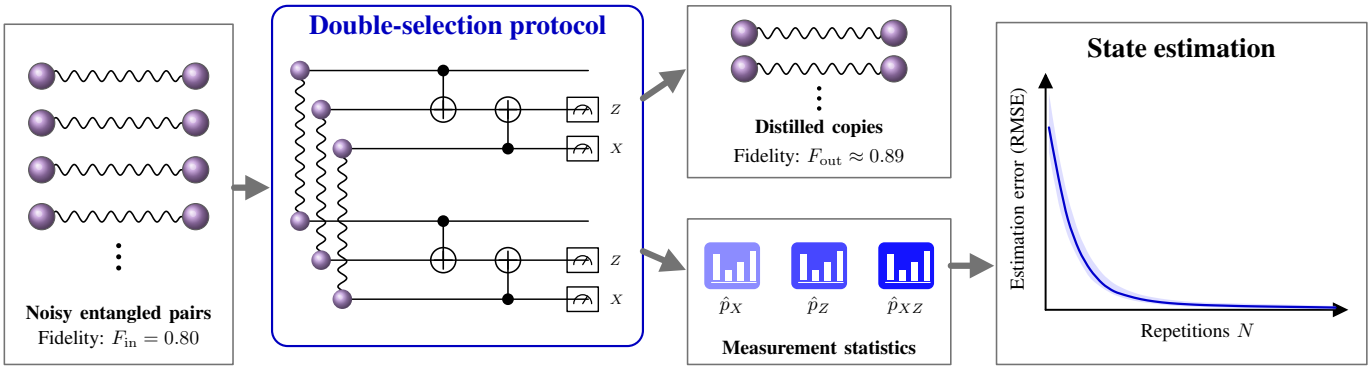


Fig. 1. Illustration of the double-selection protocol, with an example demonstrating its dual role in entanglement distillation and state estimation. Initially, Alice and Bob share independent copies of a Bell-diagonal state with fidelity 0.80 with respect to $|\Phi^+\rangle$ and equal weights on the other three Bell states. The subsystems A_i and B_i form the ith shared entangled pair. In each attempt, three pairs undergo bilateral CNOT operations, after which the second and third pairs are measured in the Z and X bases, respectively. Under ideal local operations and measurements, the first pair is retained when Alice's and Bob's outcomes coincide in both bases, yielding an output state with fidelity approximately 0.89. Meanwhile, measurement records from all attempts, including unsuccessful ones, are used to reconstruct the input Bell-diagonal state.

Together with (16) *and Lemma 1, these quantities uniquely determine the [XW: input Bell-diagonal state]. [XW: I don't like this. Here, we are in the preliminary, which is to state prior work on double selection for estimating the specifications of the input Bell state. It is US that propose to use this distimator for Pauli channel estimation, and we should state this in out methodology section IV.]*

Notice that we can *uniquely* estimate q_Z without the fidelity condition $F(\rho) > 1/2$ as the first equation in (18) is a cube root, while the other two q_X, q_Y, via square roots, rely on the $F(\rho) > 1/2$ condition.

Observation 3. *The estimation of parameter q_Z in the double-selection protocol is unique by itself and does not rely on the fidelity condition $F > \frac{1}{2}$.*

Later on, we will leverage this observation and another channel dressing technique to remove the requirement of the $F > \frac{1}{2}$ condition in our protocols.

III. Problem Formulation

Star Quantum Network.A three-link star quantum network, illustrated in Fig. 2, is represented by a graph $\mathcal{G} := (\mathcal{N}, \mathcal{L})$, where $\mathcal{N}$ denotes the set of nodes and $\mathcal{L}$ denotes the set of quantum links connecting them. We model each quantum link as a Pauli channel. More general quantum channels can be converted into effective Pauli channels through Pauli twirling [7, 8].

Here, C is the central node and N_1, N_2, N_3 are peripheral nodes, as illustrated in Fig. 2. Each peripheral node N_ℓ is connected to the central node C through one quantum link which is described by a Pauli channel with unknown channel parameters,

$$\mathcal{P}_\ell(\sigma) = A_\ell \sigma + B_\ell X\sigma X + C_\ell Y\sigma Y + D_\ell Z\sigma Z, \forall \ell \in \{1,2,3\}, \tag{19}$$

where parameters $A_\ell, B_\ell, C_\ell, D_\ell \geq 0$ and $A_\ell + B_\ell + C_\ell + D_\ell = 1$. Its nontrivial PTM eigenvalues $q_{X,\ell}, q_{Y,\ell}, q_{Z,\ell}$ are defined in (14). Each link is therefore completely characterized by either $(A_\ell, B_\ell, C_\ell, D_\ell)$ or $(q_{X,\ell}, q_{Y,\ell}, q_{Z,\ell})$. We impose the following nonzero condition on the link channels to simplify the reconstruction analysis by excluding degenerate cases[1].

Assumption 4 (Nonzero PTM eigenvalues)**.** *For every link $\ell \in \{1,2,3\}$, all nontrivial PTM eigenvalues are nonzero:*

$$q_{P,\ell} \neq 0, \qquad P \in \{X,Y,Z\}, \quad \ell \in \{1,2,3\}. \tag{20}$$

Peripheral Monitors and End-to-End Access. The peripheral nodes N_1, N_2, N_3, referred to as *monitors* , provide the state preparation and measurement (SPAM) capabilities used for tomography. The central node C can route quantum systems, perform prescribed local operations, including quantum gates S, H, and controlled-NOT (CNOT) gates, and discard auxiliary qubits. However, its SPAM capabilities, if present, are inaccessible to the tomography procedure.

This access model follows the end-to-end monitoring framework in the classical network tomography literature and is also adopted in quantum network tomography [12, 13]. It also reflects practical scenarios in which peripheral nodes are accessible to users, whereas the central node is managed by a service provider that permits prescribed operations but does not grant users access to its internal SPAM capabilities or measurement records.

End-to-End Distillation. Under the access constraints described above, we consider only end-to-end (path-level) entanglement distillation between peripheral monitoring nodes. The local measurements required at both endpoints by standard

[1]This condition is sufficient for the proposed reconstruction but is not necessary for identifiability. If $q_{X,\ell} = q_{Y,\ell} = q_{Z,\ell} = 0$, then $\mathcal{P}_\ell$ is completely depolarizing and $(\mathcal{I} \otimes \mathcal{P}_\ell)(\rho) = I \otimes I/4$ for every Bell-diagonal state ρ [9]. Such a link erases all correlations and can be treated as a failed link using existing link-failure identification methods [10, 11]. If only some PTM eigenvalues vanish, the Clifford dressings in §IV-C can permute the zero and nonzero entries. Alternative configurations may then recover the remaining parameters whenever sufficiently many nonsingular product equations are available. A complete analysis of these degenerate cases is beyond the scope of this paper.

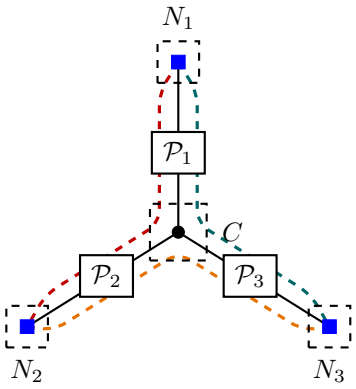


Fig. 2. Three-link star network with peripheral monitors N_1, N_2, N_3 (blue squares) and a central node C (black circle). Each link ℓ is modeled by an unknown Pauli channel $\mathcal{P}_\ell$. The monitors provide SPAM capabilities, while C performs prescribed operations without supplying measurement outcomes to the tomography procedure. The red, teal, and orange dashed lines represent the three end-to-end paths N_1-C-N_2, N_1-C-N_3, and N_2-C-N_3, respectively.

distillation protocols [14, 15] can then be performed at the monitors, with their outcomes accessible to the tomography procedure. Link-level distillation would instead involve measurements at C, whose outcomes are unavailable to users. Even if the service provider performs such distillation internally, we exclude it from our model and use only measurement data obtained from path-level distillation.

Specifically, we apply the double-selection protocol described in § II-C to entangled pairs shared between peripheral monitoring nodes. Besides conditionally retaining an output pair, the protocol provides measurement statistics for estimating the Pauli correlations of the input Bell-diagonal state.

We use these statistics in two reconstruction schemes. Under the fidelity condition $F(\rho) > 1/2$, Lemma 2 allows all three nontrivial Pauli correlations to be estimated using Eq. (18). Alternatively, Lemma 3 allows the Z correlation to be estimated from p_Z alone without this fidelity condition. Suitable Clifford dressings extend the latter approach to the X and Y components. In both schemes, the estimated end-to-end correlations are then combined to infer the individual link-level channel parameters.

The following assumption ensures the applicability of Lemma 2 to the first scheme. It is not required by the second scheme, which is based on Lemma 3.

Assumption 5 (Fidelity condition for the first reconstruction scheme)**.** *The Bell-diagonal states* ρ_{12}, ρ_{13}, $\rho_{23}^{(0)}$, *and* $\rho_{23}^{(1)}$, *defined in* (22), (23), *and* (26), *each have fidelity greater than* $1/2$ *with respect to* $|\Phi^+\rangle$.

Objective. Given the three-link star network $\mathcal{G}$, our objective is to design end-to-end protocols that use distillation measurement outcomes to estimate all link-level channel parameters $q_{P,\ell}$, for $\ell \in \{1,2,3\}$ and $P \in \{X,Y,Z\}$. We quantify [XW: the estimation accuracy using the root mean square error (RMSE)],

$$\epsilon_{\text{est}} := \sqrt{\frac{1}{9}\sum_{\ell=1}^{3}\sum_{P\in\{X,Y,Z\}}\left(\widehat{q}_{P,\ell} - q_{P,\ell}\right)^2}, \tag{21}$$

where $\widehat{q}_{P,\ell}$ denotes the estimate of $q_{P,\ell}$. Our objective is to minimize this estimation error for a given measurement budget.

Remark 6 (Extension to general network topologies). Due to space limitations, this work focuses on the three-link star topology, which provides a fundamental setting in which individual link parameters can be reconstructed from end-to-end observations without access to the central node's measurement outcomes.

The proposed framework can be naturally extended to more general quantum-network topologies without degree-two nodes based on two observations. First, any quantum network without degree-two nodes can be viewed as being composed of three-link star substructures. Second, for tomography purposes, two consecutive Pauli channels can be regarded as a single effective Pauli channel.

IV. Distillation-Assisted Network Tomography

In this section, we develop a distillation-assisted tomography protocol for the three-link star network. §IV-A first introduces direct pairwise distillation and explains its identifiability limitation. §IV-B then presents a modified `Mergecast` construction that provides correlations[XW: don't use correlations anymore, use "estimation" or "protocol", etc.] involving all three links. §IV-C uses Clifford dressing to obtain the additional estimates needed for complete reconstruction. Finally, §IV-D combines these measurements to construct an estimator and establish the identifiability of all three channels in the star network.

A. `Unicast`-Assisted Distillation

We begin with direct `Unicast` [XW: cite]transmission between peripheral monitors, [XW: where `Unicast` refers to single-path probe between one sender monitor and one receiver monitor (in the peripheral) [16].] The three-link star network contains three two-link paths, each for one pair of monitors as in Fig. 2. Consider the path N_1-C-N_2 shown in Fig. 3. The `Unicast` procedure is as follows: monitor N_1 prepares [XW: a Bell pair] $|\Phi^+\rangle$, retains one qubit locally, and sends the other to N_2 through the Pauli channel $\mathcal{P}_1$, the central node C, and the Pauli channel $\mathcal{P}_2$. The resulting state shared by monitor N_1 and N_2 is Bell diagonal, as shown in Fig. 3.

Applying the same procedure along the all three paths, including the other two N_2-C-N_3 and N_1-C-N_3, gives the quantum states with restricted Pauli–Liouville representations as follows,

$$\begin{aligned}
|\rho_{12}\rangle\rangle_{\text{B}} &= \begin{bmatrix}1 & q_{X,1}q_{X,2} & -q_{Y,1}q_{Y,2} & q_{Z,1}q_{Z,2}\end{bmatrix}^T,\\
|\rho_{13}\rangle\rangle_{\text{B}} &= \begin{bmatrix}1 & q_{X,1}q_{X,3} & -q_{Y,1}q_{Y,3} & q_{Z,1}q_{Z,3}\end{bmatrix}^T,\\
|\rho_{23}\rangle\rangle_{\text{B}} &= \begin{bmatrix}1 & q_{X,2}q_{X,3} & -q_{Y,2}q_{Y,3} & q_{Z,2}q_{Z,3}\end{bmatrix}^T.
\end{aligned} \tag{22}$$

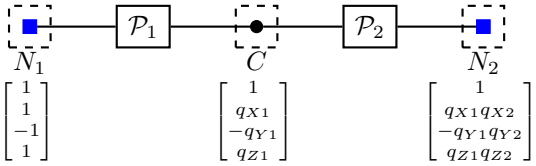


Fig. 3. Two-link path $N_1{-}C{-}N_2$ through the central node C. The column vectors below the nodes are the restricted Pauli–Liouville representation of the shared state at the corresponding transmission stage. By symmetry, the same protocol can be applied to the other two two-link paths, $N_1 - C - N_3$ and $N_2 - C - N_3$.

For each path, the two monitors independently generate three copies of the shared state [XW: via the `Unicast` transmission above] and then perform the double-selection protocol in §II-C.

Whenever the source state for distillation in (22) satisfies the fidelity condition of Lemma 2, the double-selection statistics in (17) *uniquely estimate* its nontrivial Pauli correlations (e.g., the last three entries of $|\rho_{12}\rangle\rangle_{\mathrm{B}}$: $\widehat{q_{X,1}q_{X,2}}$, $\widehat{q_{Y,1}q_{Y,2}}$ and $\widehat{q_{Z,1}q_{Z,2}}$). Thus, applying this procedure to all three paths estimate any $\widehat{q_{P,\ell}q_{P,\ell'}}$ for any $P \in \{X,Y,Z\}$ and $(\ell,\ell') \in \{(1,2),(1,3),(2,3)\}$.

a) Identifiability challenge: For each $P \in \{X,Y,Z\}$, the double-selection statistics yield estimates of the pairwise products $q_{P,1}q_{P,2}$, $q_{P,1}q_{P,3}$, and $q_{P,2}q_{P,3}$, each involving an even number of factors. From these three estimates, one can, for instance, estimate the parameter $q_{P,1}$ as $\pm\sqrt{\frac{\widehat{q_{P,1}q_{P,2}}\times\widehat{q_{P,1}q_{P,3}}}{\widehat{q_{P,2}q_{P,3}}}}$, with a sign ambiguity. That is, there are two possible sign choices for each Pauli component, across all three components giving up to eight choices in total. This ambiguity suggests that this protocol alone cannot uniquely identify all channels.

To resolve this sign ambiguity, we introduce a `Mergecast` protocol in the following subsection, which generates estimates involving all three links (i.e., products of an odd number of factors). By combining the estimates with the ones in `Unicast`, we resolve this sign ambiguity and uniquely estimate the channel parameters.

B. `Mergecast`-Assisted Distillation

Fig. 4 illustrates the `Mergecast` protocol [12], which introduces correlations involving all three links to resolve the sign ambiguity. We apply this protocol to the path $N_2{-}C{-}N_3$ and then use the double-selection protocol to estimate the resulting Bell-diagonal state. The procedure is as follows.

1) Node N_2 prepares $|\Phi^+\rangle$, retains one qubit, and sends the other to C through $\mathcal{P}_2$. Meanwhile, N_1 sends an auxiliary qubit in $|0\rangle$ to C through $\mathcal{P}_1$.
2) Node C applies a CNOT with the auxiliary qubit as the control and the Bell-pair qubit as the target. It then traces out (i.e., discard) the control and forwards the target to N_3 through $\mathcal{P}_3$.
3) After generating three independent copies from repeating steps 1) and 2), node N_2 and N_3 perform double-selection protocol.

Before distillation, each source state shared by N_2 and N_3 is

$$\left|\rho_{23}^{(0)}\right\rangle\!\right\rangle_{\mathrm{B}} = \begin{bmatrix}1 & q_{X,2}q_{X,3} & -q_{Z,1}q_{Y,2}q_{Y,3} & q_{Z,1}q_{Z,2}q_{Z,3}\end{bmatrix}^{\mathsf{T}}. \tag{23}$$

By Lemma 2, the double-selection statistics yield estimates of the nontrivial Pauli correlations of $\rho_{23}^{(0)}$. For example, combining the resulting estimate of $q_{Z,1}q_{Z,2}q_{Z,3}$ with the previously obtained estimate of $q_{Z,1}q_{Z,2}$ gives $\widehat{q}_{Z,3} := \frac{\widehat{q_{Z,1}q_{Z,2}q_{Z,3}}}{\widehat{q_{Z,1}q_{Z,2}}}$. This illustrates how combining the estimates from the two protocols resolves the sign ambiguity. Further details are presented in § IV-D.

To recover the remaining parameters $q_{X,\ell}$ and $q_{Y,\ell}$, we need to access correlations in other Pauli bases. We achieve this through channel dressing, as described below.

Remark 7 (Fidelity inefficiency of the `Mergecast` protocol). Although this `Mergecast` operation may reduce the fidelity of the final source state in (23) shared by node N_2 and N_3, it provides information that cannot be obtained from `Unicast` and resolve the sign ambiguity issue. In other words, it trades some end-to-end fidelity for the identifiability.

C. Clifford-Dressed Pauli Channels

We adapt the channel-dressing technique [XW: of Wang et al.] [12] to the `Mergecast` protocol. Specifically, [XW: in Fig. 4,] we apply different Clifford dressings to $\mathcal{P}_2$ and $\mathcal{P}_3$, while leaving $\mathcal{P}_1$ unchanged. [XW: Dressing the channels is equivalent to change input and output bases of the channels, providing the statistics for reconstructing other channel parameters.]

The dressing uses the Hadamard gate H and phase gate S. Both are Clifford gates, which bijectively map Pauli operators to Pauli operators up to a sign under conjugation [17]:

$$\begin{aligned} HIH^\dagger &= I, & SIS^\dagger &= I,\\ HXH^\dagger &= Z, & SXS^\dagger &= Y,\\ HYH^\dagger &= -Y, & SYS^\dagger &= -X,\\ HZH^\dagger &= X, & SZS^\dagger &= Z. \end{aligned} \tag{24}$$

Ignoring signs, H exchanges X and Z, while S exchanges X and Y. Together, these exchanges generate all permutations of $\{X,Y,Z\}$. When a Pauli channel is dressed by a Clifford gate and its Hermitian conjugate, the signs cancel, so the dressing simply permutes its PTM eigenvalues. For a unitary U, define $\mathcal{U}_U(\sigma) := U\sigma U^\dagger$. We use the Clifford gates SHS and S to define

$$\begin{aligned} \mathcal{P}_2^{(SHS)} &:= \mathcal{U}_{SHS} \circ \mathcal{P}_2 \circ \mathcal{U}_{(SHS)^\dagger},\\ \mathcal{P}_3^{(S)} &:= \mathcal{U}_S \circ \mathcal{P}_3 \circ \mathcal{U}_{S^\dagger}. \end{aligned} \tag{25}$$

These dressings exchange the Y and Z eigenvalues of $\mathcal{P}_2$, and the X and Y eigenvalues of $\mathcal{P}_3$, respectively, as illustrated in Fig. 5.

Replacing $\mathcal{P}_2$ and $\mathcal{P}_3$ with these dressed channels in the `Mergecast`-based distillation protocol produces

$$\left|\rho_{23}^{(1)}\right\rangle\!\right\rangle_{\mathrm{B}} = \begin{bmatrix}1 & q_{X,2}q_{Y,3} & -q_{Z,1}q_{Z,2}q_{X,3} & q_{Z,1}q_{Y,2}q_{Z,3}\end{bmatrix}^{\mathsf{T}}. \tag{26}$$

if space is limited, we can omit this detail.

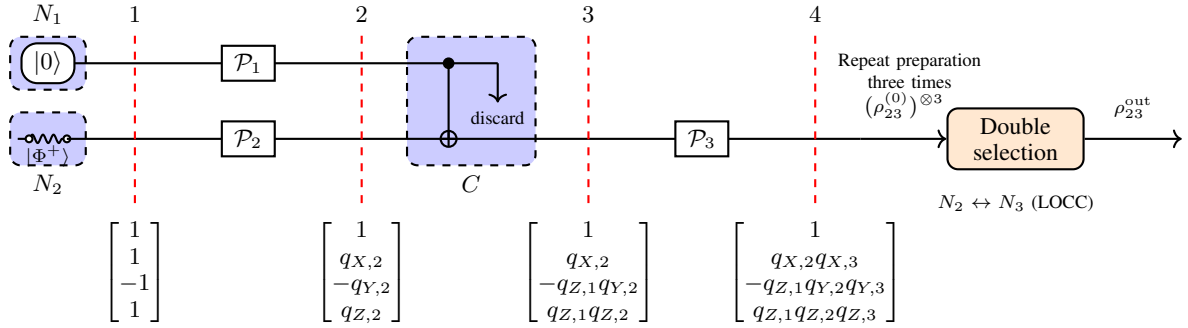


Fig. 4. `Mergecast` preparation followed by double-selection protocol. Slices 1–4 illustrate the entanglement-generation procedure, adapted from the `Mergecast` protocol in [12]. The column vectors give the restricted Pauli–Liouville representations of the reduced two-qubit state. The preparation is repeated independently three times, after which monitors N_2 and N_3 perform double-selection protocol to distill a retained entangled pair and estimate the input-state parameters from the measurement statistics.

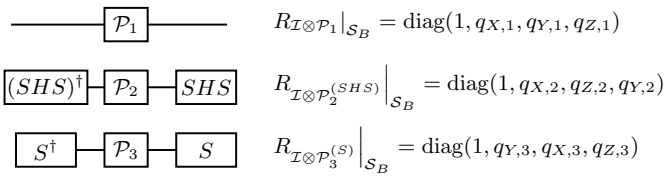


Fig. 5. Link-dependent Clifford dressing used in the modified `Mergecast` protocol. Each channel $\mathcal{P}_\ell$ is preceded by $U_\ell^\dagger$ and followed by U_ℓ, with $U_1 = I$, $U_2 = SHS$, and $U_3 = S$. Thus, $\mathcal{P}_1$ remains unchanged, while the PTM eigenvalues of $\mathcal{P}_2$ and $\mathcal{P}_3$ are permuted as $(q_{X,2}, q_{Y,2}, q_{Z,2}) \mapsto (q_{X,2}, q_{Z,2}, q_{Y,2})$ and $(q_{X,3}, q_{Y,3}, q_{Z,3}) \mapsto (q_{Y,3}, q_{X,3}, q_{Z,3})$, respectively. These dressings allow the `Mergecast` protocol to access the mixed products $q_{Z,1}q_{Z,2}q_{X,3}$ and $q_{Z,1}q_{Y,2}q_{Z,3}$, which provide the additional information needed to resolve the remaining sign ambiguities.

[XW: how about $|\rho_{\text{merge},23}\rangle\rangle_{\text{B}}$ for $\left|\rho_{23}^{(0)}\right\rangle\!\right\rangle_{\text{B}}$, and $|\rho_{\text{dress},23}\rangle\rangle_{\text{B}}$ for $\left|\rho_{23}^{(1)}\right\rangle\!\right\rangle_{\text{B}}$. The superscript is going to be used for other things later.] By Lemma 2, the double-selection statistics recovers the nontrivial Pauli [XW: coefficients] of $\rho_{23}^{(1)}$. In particular, the last two [XW: coefficients] provide the information needed to determine $q_{X,3}$ and $q_{Y,2}$. Together with the direct and undressed `Mergecast` measurements, they enable the complete channel reconstruction described next.

D. Link-Parameter Reconstruction

a) Estimator with [XW: Assumption 5 for source-state fidelity]: We now combine the estimates obtained from the four configurations to reconstruct the individual channel parameters.

By Lemma 2, the double-selection statistics of the directly generated states ρ_{12} and ρ_{13} yield estimators $\widehat{\pi}_{12}^{(P)}$ and $\widehat{\pi}_{13}^{(P)}$ of the pairwise products [XW: defined as follows,]

$$\pi_{12}^{(P)} \coloneqq q_{P,1}q_{P,2},\ \pi_{13}^{(P)} \coloneqq q_{P,1}q_{P,3},\ P \in \{X, Y, Z\}. \tag{27}$$

Similarly, the double-selection statistics of the undressed `Mergecast` state $\rho_{23}^{(0)}$ yield an estimator $\widehat{\tau}_Z$ of the three-link correlation

$$\tau_Z \coloneqq q_{Z,1}q_{Z,2}q_{Z,3}.$$

Combining these estimates, we define

$$\widehat{q}_{Z,1} \coloneqq \frac{\widehat{\pi}_{12}^{(Z)}\widehat{\pi}_{13}^{(Z)}}{\widehat{\tau}_Z}, \widehat{q}_{Z,2} \coloneqq \frac{\widehat{\tau}_Z}{\widehat{\pi}_{13}^{(Z)}}, \widehat{q}_{Z,3} \coloneqq \frac{\widehat{\tau}_Z}{\widehat{\pi}_{12}^{(Z)}}. \tag{28}$$

To estimate the remaining parameters, we use the double-selection statistics of the dressed `Mergecast` state $\rho_{23}^{(1)}$. These statistics yield estimators $\widehat{\mu}_X$ and $\widehat{\mu}_Y$ of

$$\mu_X \coloneqq q_{Z,1}q_{Z,2}q_{X,3}, \qquad \mu_Y \coloneqq q_{Z,1}q_{Y,2}q_{Z,3}.$$

Dividing these estimates by the corresponding estimates gives

$$\widehat{q}_{X,3} \coloneqq \frac{\widehat{\mu}_X}{\widehat{\pi}_{12}^{(Z)}}, \qquad \widehat{q}_{Y,2} \coloneqq \frac{\widehat{\mu}_Y}{\widehat{\pi}_{13}^{(Z)}}. \tag{29}$$

These expressions provide estimates of both the magnitude and sign of one parameter in each of the X and Y sectors. The remaining parameters are then estimated sequentially as

$$\begin{aligned} \widehat{q}_{X,1} &\coloneqq \frac{\widehat{\pi}_{13}^{(X)}}{\widehat{q}_{X,3}}, \quad \widehat{q}_{X,2} \coloneqq \frac{\widehat{\pi}_{12}^{(X)}}{\widehat{q}_{X,1}}, \\ \widehat{q}_{Y,1} &\coloneqq \frac{\widehat{\pi}_{12}^{(Y)}}{\widehat{q}_{Y,2}}, \quad \widehat{q}_{Y,3} \coloneqq \frac{\widehat{\pi}_{13}^{(Y)}}{\widehat{q}_{Y,1}}. \end{aligned} \tag{30}$$

Finally, applying the transformation in Lemma 1 gives estimators of the Pauli-error probabilities of each channel:

$$\begin{bmatrix} \widehat{A}_\ell \\ \widehat{B}_\ell \\ \widehat{C}_\ell \\ \widehat{D}_\ell \end{bmatrix} \coloneqq W^T \begin{bmatrix} 1 \\ \widehat{q}_{X,\ell} \\ -\widehat{q}_{Y,\ell} \\ \widehat{q}_{Z,\ell} \end{bmatrix}, \qquad \ell \in \{1, 2, 3\}. \tag{31}$$

To summarize, equations (28)–(30) give the estimates of the link parameters, while (31) gives the corresponding estimates of the Pauli-error probabilities. The exact reconstruction above establishes the following identifiability result.

Theorem 8 (Reconstruction with a fidelity assumption)**.** *Under Assumptions 4 , all nine link-level PTM eigenvalues, and hence all three Pauli channels, are uniquely identifiable from double-selection statistics [XW: via (28)–(30)] by combining the* `Unicast` *and* `Mergecast` *protocols with link-dependent Clifford dressing.*

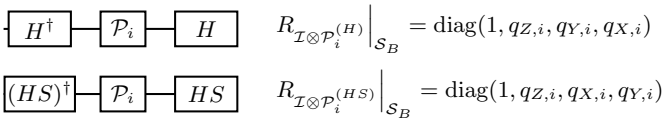


Fig. 6. Clifford dressing of the Pauli channel $\mathcal{P}_i$ and the resulting permutations of its PTM eigenvalues. The upper and lower circuits implement the H and HS dressings, respectively, with the corresponding restricted PTMs shown on the right. These dressings give $q_{Z,i}^{(H)} = q_{X,i}$ and $q_{Z,i}^{(HS)} = q_{Y,i}$, allowing the original X and Y parameters to be estimated through the effective Z parameter.

b) Estimator without Assumption 5: We can still estimate $q_{Z,i}$ using (28) without Assumption 5. Indeed, by [XW: Observation] 3, all required estimates [XW: in] the corresponding Z-basis [XW: can be obtained uniquely] without imposing the fidelity condition.

To estimate $q_{X,\ell}$ and $q_{Y,\ell}$, we apply each of the Clifford dressings shown in Fig. 6 uniformly to all three channels in the network. This differs from the previous method, which applies different dressings to different channels [XW: in `Mergecast`]. The H and HS dressings map $q_{X,\ell}$ and $q_{Y,\ell}$, respectively, to the effective Z eigenvalues of the dressed channels:

$$q_{Z,\ell}^{(H)} = q_{X,\ell}, \qquad q_{Z,\ell}^{(HS)} = q_{Y,\ell}, \qquad \forall \ell \in \{1,2,3\}. \tag{32}$$

For each dressing, we apply the same transformation to all three links and repeat the `Unicast` and undressed `Mergecast` procedures with the dressed channels. Lemma 3 then yields estimates of the corresponding pairwise and three-link products from the p_Z statistics. Substituting these estimates into (28), with Z replaced by X or Y, gives estimators of $q_{X,i}$ and $q_{Y,i}$. This establishes the following reconstruction result. [XW: give the specific estimator in detail here, and refer to the estiamtors in the following theorem.]

Theorem 9 (Reconstruction without a fidelity assumption)**.** *Even without Assumption 5, all link-level PTM eigenvalues of these three Pauli channels in the star network remain uniquely identifiable from double-selection statistics by combining Clifford dressing with the* `Unicast` *and* `Mergecast` *protocols.*

[XW: add a discussion to compare this and the above theorem, espeically on the efficiency.] [XW: defer the following to our main result section, and highlight as one key contribution. Also, we don't need a lemma to highlight this. Use the environment: Observation.] When this assumption is not imposed, Lemma 3 still allows q_Z to be reconstructed from p_Z alone, but leaves the information contained in p_X and p_{XZ} unused. This tradeoff between fidelity requirements and the use of measurement statistics motivates two distinct reconstruction schemes for the QNT problem, based on the two lemmas, respectively.

V. Numerical Experiment

References

[1] J. Miguel-Ramiro and W. Dür, "Efficient entanglement purification protocols for ¡mml:math xmlns:mml="http://www.w3.org/1998/math/mathml"¿ ¡mml:mi¿d¡/mml:mi¿ ¡/mml:math¿ -level systems," *Physical Review A*, vol. 98, no. 4, Oct. 2018. [Online]. Available: http://dx.doi.org/10.1103/PhysRevA.98.042309

[2] J. Lin, J. J. Wallman, I. Hincks, and R. Laflamme, "Independent state and measurement characterization for quantum computers," *Physical Review Research*, vol. 3, no. 3, p. 033285, Sep. 2021.

[3] R. Horodecki and M. Horodecki, "Information-theoretic aspects of inseparability of mixed states," *Phys. Rev. A*, vol. 54, pp. 1838–1843, Sep 1996. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.54.1838

[4] S. Luo, "Quantum discord for two-qubit systems," *Phys. Rev. A*, vol. 77, p. 042303, Apr 2008. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.77.042303

[5] J. C. A. Casapao, A. G. Maity, N. Benchasattabuse, M. Hajdušek, A. Soeda, R. Van Meter, and D. Elkouss, "A double selection entanglement distillation-based state estimator," in *2025 IEEE International Conference on Quantum Computing and Engineering (QCE)*. IEEE, Aug. 2025, p. 1089–1097. [Online]. Available: http://dx.doi.org/10.1109/QCE65121.2025.00122

[6] K. Fujii and K. Yamamoto, "Entanglement purification with double selection," *Phys. Rev. A*, vol. 80, p. 042308, Oct 2009. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.80.042308

[7] W. Dür, M. Hein, J. I. Cirac, and H.-J. Briegel, "Standard forms of noisy quantum operations via depolarization," *Phys. Rev. A*, vol. 72, p. 052326, Nov 2005. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.72.052326

[8] C. H. Bennett, D. P. DiVincenzo, J. A. Smolin, and W. K. Wootters, "Mixed-state entanglement and quantum error correction," *Phys. Rev. A*, vol. 54, pp. 3824–3851, Nov 1996. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevA.54.3824

[9] D. Gottesman, *Surviving as a Quantum Computer in a Classical World*. Unpublished manuscript, 2026. [Online]. Available: https://www.cs.umd.edu/~dgottesm/QECCbook-2026.pdf

[10] S. Ahuja, S. Ramasubramanian, and M. Krunz, "Srlg failure localization in all-optical networks using monitoring cycles and paths," 05 2008, pp. 700 – 708.

[11] H. X. Nguyen and P. Thiran, "Active measurement for multiple link failures diagnosis in ip networks," in *Passive and Active Network Measurement*, C. Barakat and I. Pratt, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2004, pp. 185–194.

[12] X. Wang, M. G. D. Andrade, G. Avis, Y. zhen Janice Chen, M. Hajiesmaili, and D. Towsley, "Quantum network tomography for general topology with spam errors," 2025. [Online]. Available: https://arxiv.org/abs/2511.01074

[13] T. He, L. Ma, A. Swami, and D. Towsley, *Network Tomography: Identifiability, Measurement Design, and*

Network State Inference. Cambridge University Press, 2021.

[14] C. H. Bennett, G. Brassard, S. Popescu, B. Schumacher, J. A. Smolin, and W. K. Wootters, "Purification of noisy entanglement and faithful teleportation via noisy channels," *Phys. Rev. Lett.*, vol. 76, pp. 722–725, Jan 1996. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.76.722

[15] D. Deutsch, A. Ekert, R. Jozsa, C. Macchiavello, S. Popescu, and A. Sanpera, "Quantum privacy amplification and the security of quantum cryptography over noisy channels," *Physical Review Letters*, vol. 77, no. 13, p. 2818–2821, 1996. [Online]. Available: http://dx.doi.org/10.1103/PhysRevLett.77.2818

[16] S. Pirandola, "Bounds for multi-end communication over quantum networks," *Quantum Science and Technology*, vol. 4, no. 4, p. 045006, 2019. [Online]. Available: http://dx.doi.org/10.1088/2058-9565/ab3f66

[17] D. Gottesman, "The Heisenberg representation of quantum computers," 1998. [Online]. Available: https://arxiv.org/abs/quant-ph/9807006